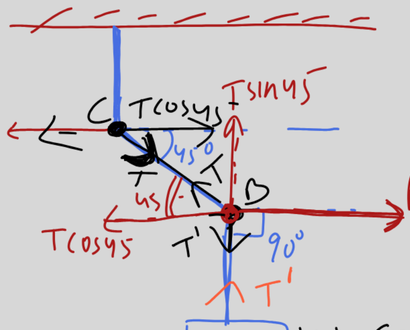
(Possible Question for Midsem)

Exercise 5.10



(a) Find Tension in the diagonal string

(b) Find F_1 & F_2 s.t system is in equilibrium

at A: $T' = W = 60N$ — (i)

F_2 at B: $T\sin45 = T' = 60$

$\Rightarrow T \times \frac{1}{\sqrt{2}} = 60$

$\Rightarrow T = 60\sqrt{2} = 84.9N$

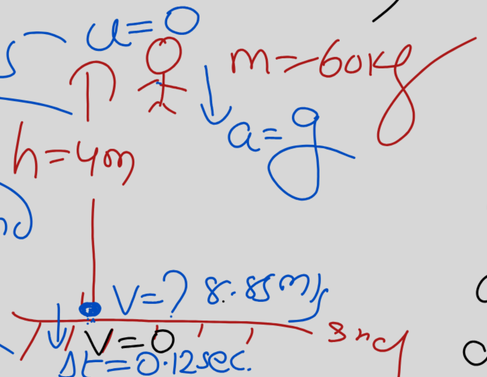
(ii) $\Rightarrow F_2 = T\cos45$

$= 60\sqrt{2} \times \frac{1}{\sqrt{2}}$

$F_1 = T\cos45 = 60N$

$= 60\sqrt{2} \times \frac{1}{\sqrt{2}} = 60N$

Exercise 5.25



V = vel of man just before striking the gnd

$V_y^2 - V_0^2 = 2gs$

$V^2 - u^2 = 2as$

$V^2 - 0 = 2(9.8)(4) \Rightarrow V = \sqrt{2 \times 9.8 \times 4} = 8.85 \text{ m/s}$

$P_i = mu = 60 \times 8.85$

$P_f = 0$

$d\vec{P} = \vec{P_f} - \vec{P_i} = 0 - 60 \times 8.85$

$dt = 0.12 \text{ sec.}$

$\vec{F} = \frac{d\vec{P}}{dt}$

$\Rightarrow \vec{F} = \frac{-60 \times 8.85}{0.12}$

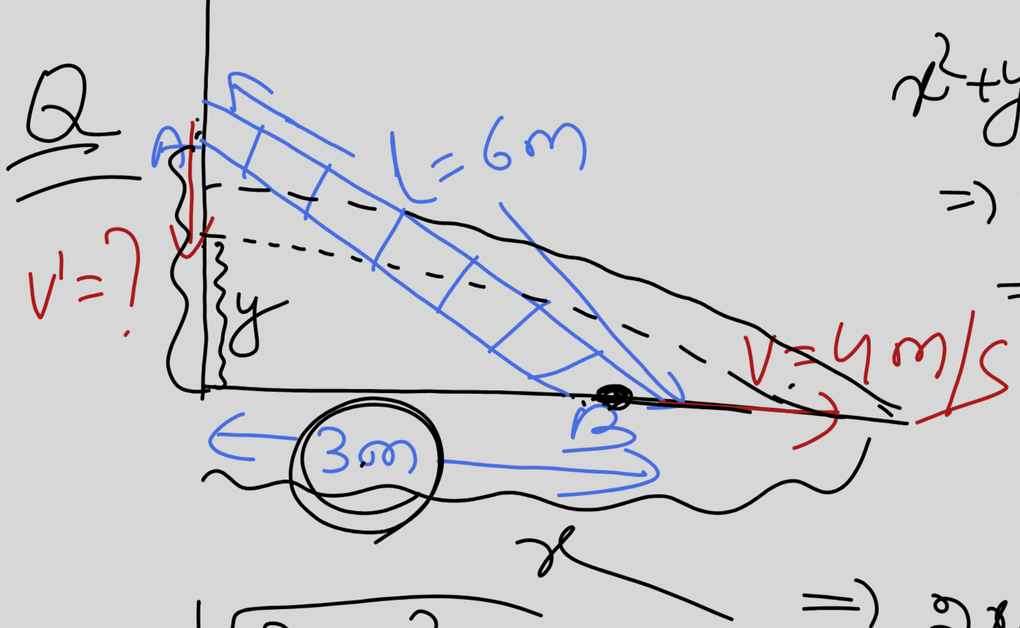
$= -4427 N$

$F = \frac{dp}{dt}$

$= \frac{-60 \times 8.85}{0.12}$

$= 885 N$

Q



$$x^2+y^2=6^2$$

$$\Rightarrow x^2+y^2=36$$

$\Rightarrow$ differentiating above w.r.t 't' i.e.

$$\frac{dx^2}{dt}+\frac{dy^2}{dt}=\frac{d}{dt}(36)$$

$$\Rightarrow 2x\left(\frac{dx}{dt}\right)+2y\left(\frac{dy}{dt}\right)=0$$

$$\Rightarrow x(v_x)+y(v_y)=0$$

$$\Rightarrow yv_y=-xv_x$$

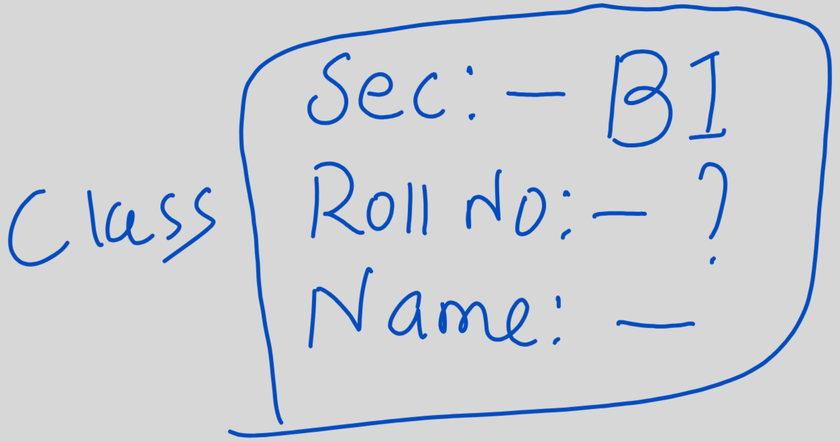
$$\Rightarrow v_y=-\left(\frac{x}{y}\right)v_x$$

$$\Rightarrow v_y=-\left(\frac{3}{3\sqrt{3}}\right)\times 4=-\frac{4}{\sqrt{3}}$$

$$y=\sqrt{6^2-3^2}$$

$$=3\sqrt{3}$$

when $x=3$


Class
Sec:- B1
Roll No:- ?
Name:-

Ex Search $x = 17$, $\{1, 5, 7, 9, 10, 13, 17, 19, 20, 21\}$
$a_1\ a_2\ a_3\ a_4\ a_5\ a_6\ a_7\ a_8\ a_9\ a_{10}$

Binary Search.

Sol $i=1 \quad j=10$

$(1<10)(T)\ m = \lfloor \frac{i+j}{2} \rfloor = \lfloor \frac{1+10}{2} \rfloor = 5$

If $x > a_5$ (T) $\quad i := m+1 = 6$

$i=6,\ j=10$

$(6<10)(T)\ m = \lfloor \frac{6+10}{2} \rfloor = 8$

If $x > a_8$ (F) $\quad j := 8$

$i=6 \quad j=8$

$(6<8)(T)\ m = \lfloor \frac{6+8}{2} \rfloor = 7$

If $x > a_7$ (F) $\quad j := 7$

$i=6,\ j=7$

$(6<7)\ m = \lfloor \frac{6+7}{2} \rfloor = 6$

$x > a_6$ (T) $\quad i = 6+1 = 7$

$i=7,\ j=7$

$(7<7)$ (F)

If $x = a_7$ (T) location $= 7$.

---

matrix

Zero-one matrix

$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$C = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$b_1 \vee b_2 = \begin{cases} 0 & \text{if } b_1 = 0 \text{ \& } b_2 = 0 \\ 1 & \text{otherwise.} \end{cases}$

$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ \& } b_2 = 1 \\ 0 & \text{otherwise.} \end{cases}$

let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$

The join of matrix A and B.

$A \vee B = [a_{ij} \vee b_{ij}]_{m \times n}$.

Meet of matrix A & B.

$A \wedge B = [a_{ij} \wedge b_{ij}]_{m \times n}$.

Ex $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Sol

$A \vee B = \begin{bmatrix} (1 \vee 1) & (0 \vee 0) \\ (0 \vee 0) & (1 \vee 0) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A \wedge B = \begin{bmatrix} (1 \wedge 1) & (0 \wedge 0) \\ (0 \wedge 0) & (1 \wedge 0) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

Boolean Product

$A \odot B$

let $A = [a_{ij}]_{m \times k}$ Zero-one matrix

$B = [b_{ij}]_{k \times n}$ Zero-one matrix

The Boolean Product of matrix A and B.

$A \odot B = [c_{ij}]_{m \times n}$

$c_{ij} = (a_{i1} \wedge a_{1j}) \vee (a_{i2} \wedge a_{2j}) \vee \cdots \vee (a_{ik} \wedge a_{kj})$

Ex $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$A \odot B = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

Ex $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$A \odot B = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) \end{bmatrix}$

$= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

nth Boolean Product of A

$A^{[n]} = \underbrace{A \odot A \odot A \odot \cdots \odot A}_{n \text{ times}}$

Ex find $A^{[n]}$, $A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$, for +ve n.

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$A^{[2]} = A \odot A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A^{[3]} = A^{[2]} \odot A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^{[4]} = A^{[3]} \odot A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^{[5]} = A^{[4]} \odot A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$A^{[6]} = A^{[5]} \odot A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \odot \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Ex

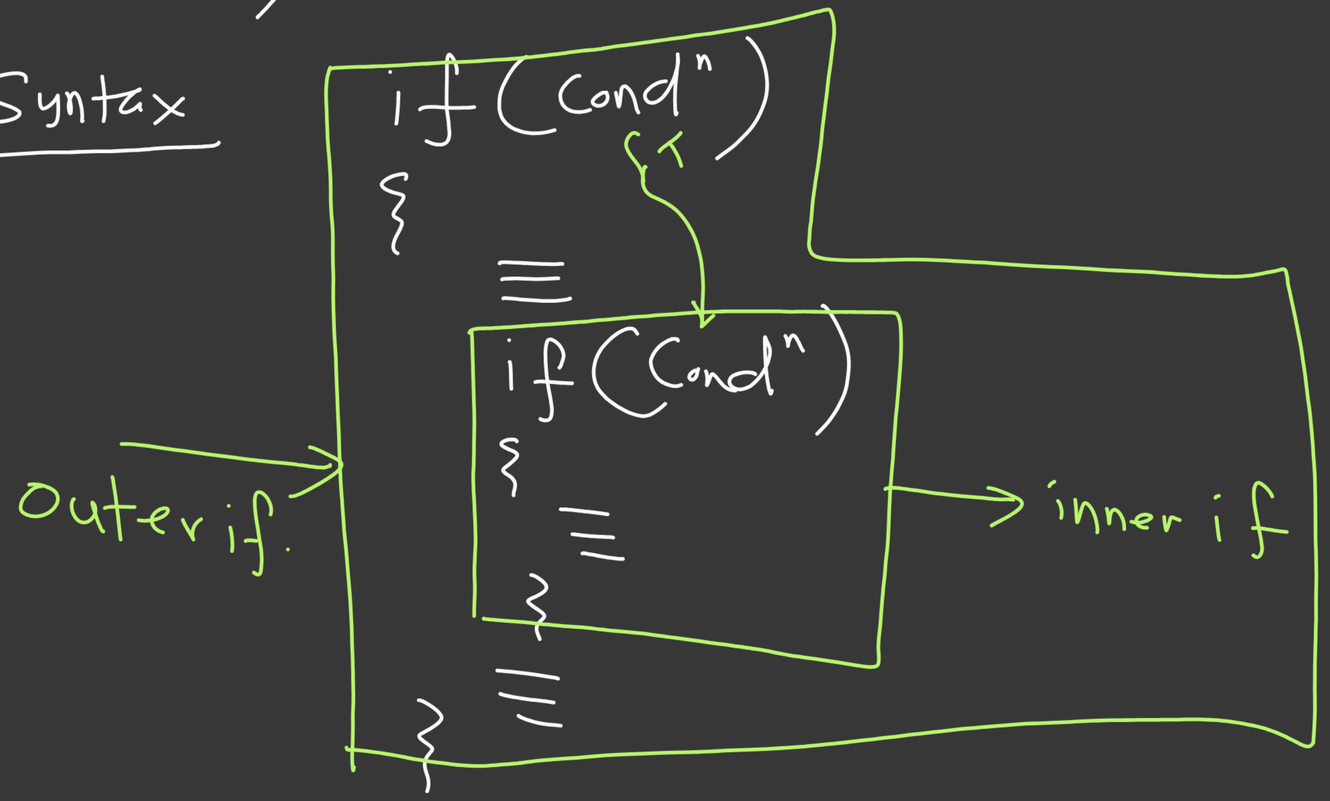
$$A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

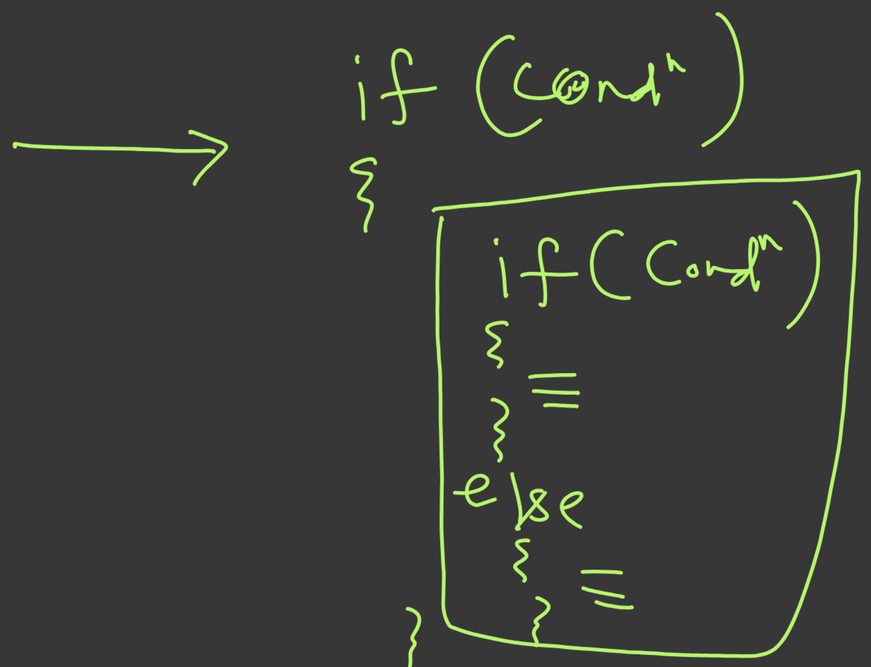
Nested if

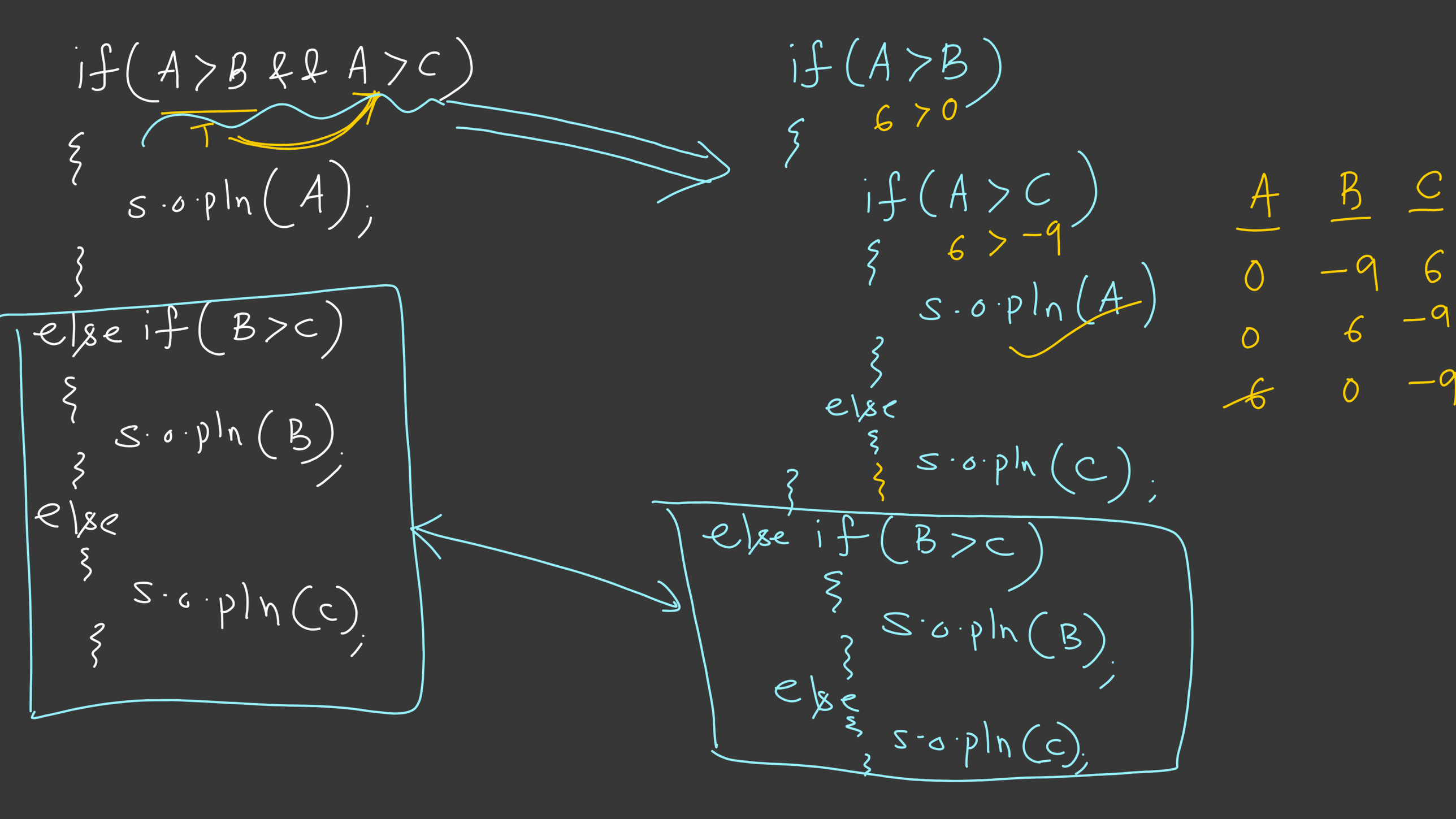
→ One if is kept inside Another.

Syntax



- The inner if is get executed only if the outer if is "true".




if(A>B && A>C)
T
{
System.out.println(A);
}
else if(B>c)
{
System.out.println(B);
}
else
{
System.out.println(c);
}
if(A>B)
6>0
{
if(A>C)
6>-9
{
System.out.println(A)
}
else
{
System.out.println(C);
}
else if(B>c)
{
System.out.println(B);
}
else
{
System.out.println(c);
}
A B C
0 -9 6
0 6 -9
6 0 -9

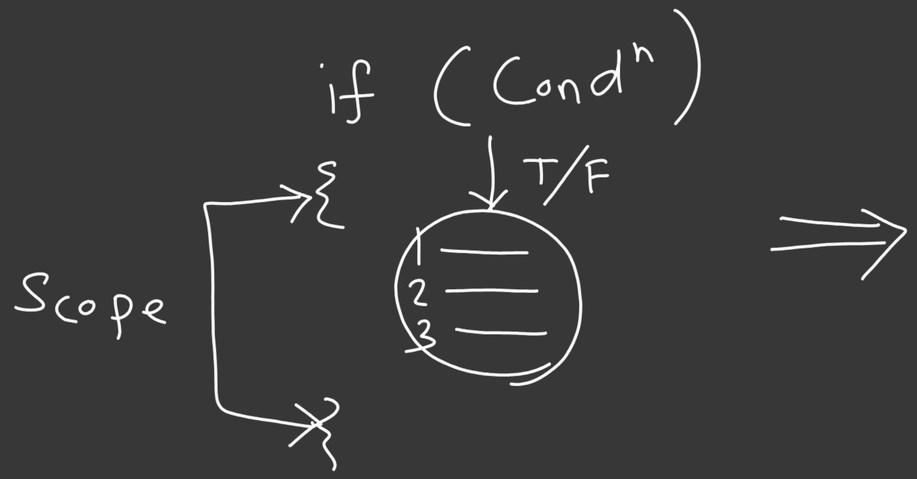
int G = Math.max(Math.max(A, B), C);

6 0, -9 0, 6

A	B	C
0	-9	6

Scope

Area that is visible / executed unde a Cond^n or Loop or method.



if (Cond^n)

1
2
3

⟶ by default one stmt is under if.

```
int a=10, b=10;
if(a!=b)
{
  s.o.pln("Hello");
  s.o.pln("Hi");
}
s.o.pln("Thank You");
```

o/p

Hello
Hi
Thank You.

o/p

Thank You.

```
int a=10, b=10;
if(a!=b)
10!=10 → F
X s.o.pln("Hello");
✓ s.o.pln("Hi");
✓ s.o.pln("Thank You");
```

o/p

Hi
Thank You.

```
int a=10, b=6;
if(a<b);
    s.o.pln("Hello");
else
    s.o.pln("Hi");
s.o.pln("Thank You");
s.o.pln("SaaYonara");
```

live & die here.

error

```
int a=10, b=6;
if (a<b)
    s.o.pln("Hello");
else
    s.o.pln("Hi");
s.o.pln("Thank You");
s.o.pln("SaaYonara");
```

error

Mis-placed else

```
int a=10, b=10;
if (a<b);
{
    s.o.pln("Hello");
    s.o.pln("Hi");
}
s.o.pln("Thank You");
```

o/p

```
Hello
Hi
Thank You.
```

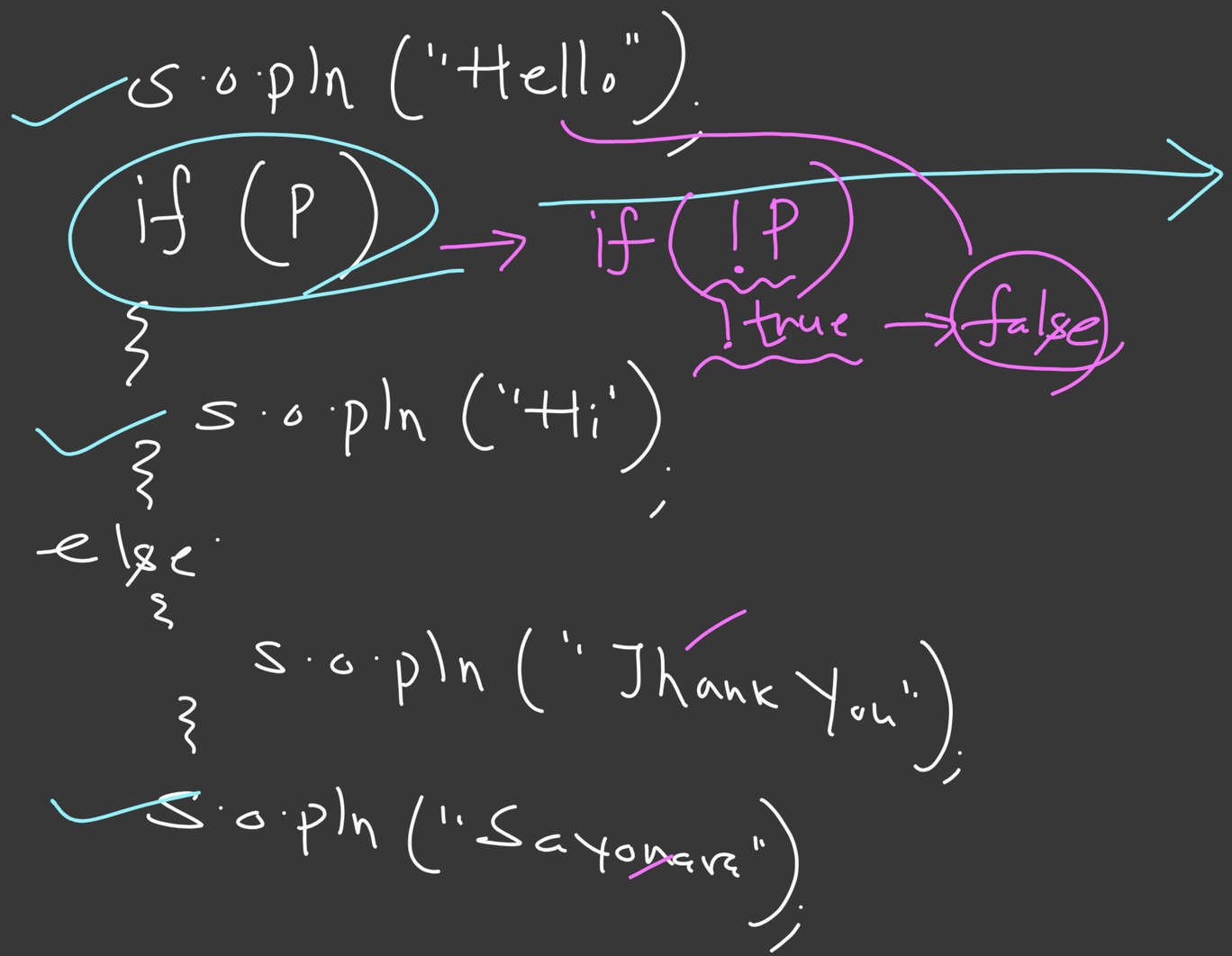
```
int a=10, b=10;
if (a!=b); T/F is terminated/Cond^n is terminated.
s.o.pln("Hello");
s.o.pln("Hi");
s.o.pln("Thank You");
```

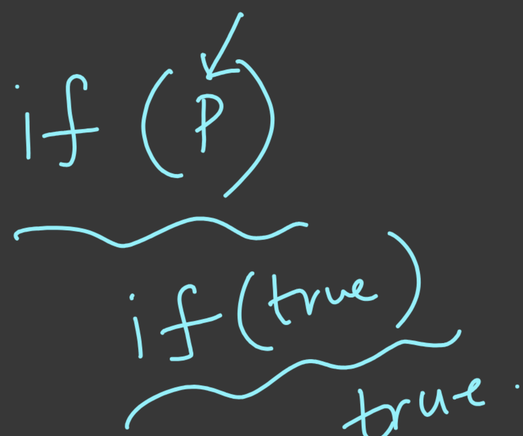
; → terminⁿ Symbol

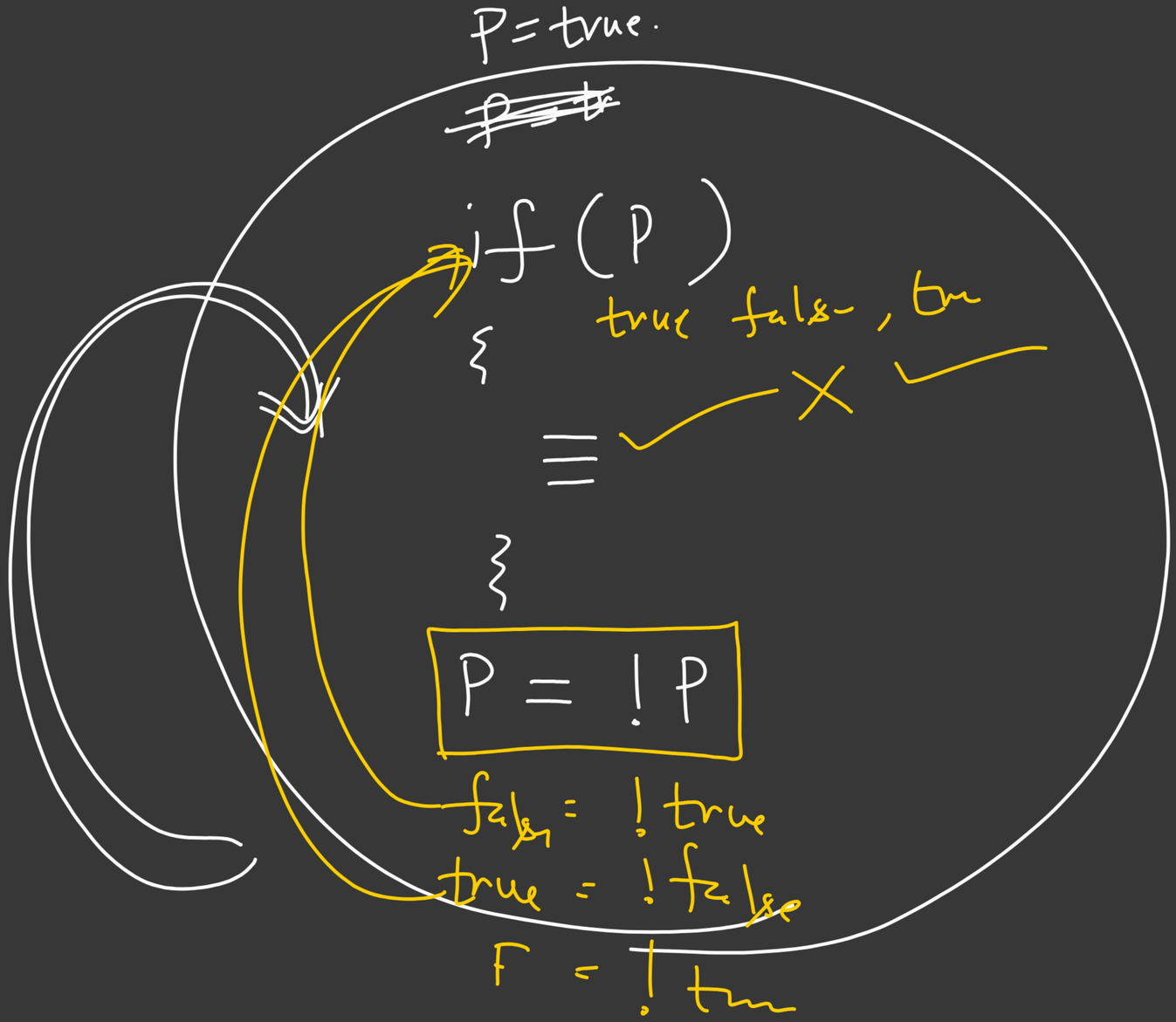
o/p

```
Hello
Hi
Thank You.
```

boolean P = true;






P = true.
if (P)
{
true false , true
X
≡
}
P = ! P
false = ! true
true = ! false
F = ! true

Q1
```
int a, b, c;
a = -13 + 2*17 - 4;
b = a*28 + 4%5 - 16;
c = a + b*3 - 2%5 - 14;
s.o.pln(a + " " + b + " " + c);
```

Q2
```
int x = -9;
int y = x++ + ++x + ++x;
int z = --y + x++ + y++;
int p = z++ - (z%10) + (p=z);
s.o.pln(x + " " + y + " " + z + " " + p);
```

Q3
```
int a = -10, b = -35;
int c = a ^ b;
int d = c & b;
s.o.pln(a + " " + b + " " + c + " " + d);
```

Q4
```
int a = -2 + 21*3 - 9/2;
int b = 44 - 12*4 + 37/8 - 3;
boolean p = (++a > 71 && --b > 100)
s.o.pln(p + " " + a + " " + b);
```